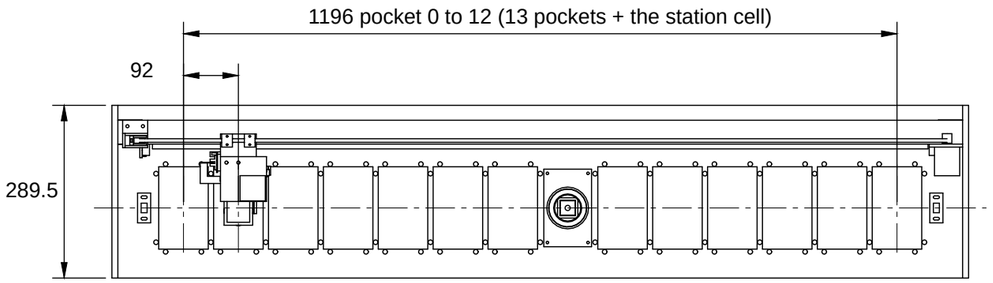
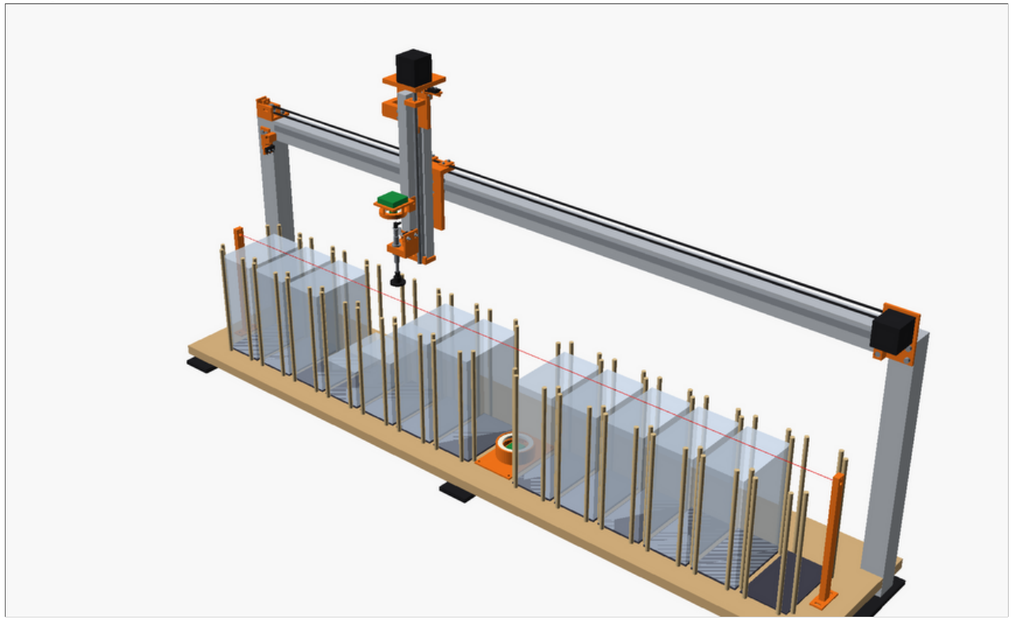
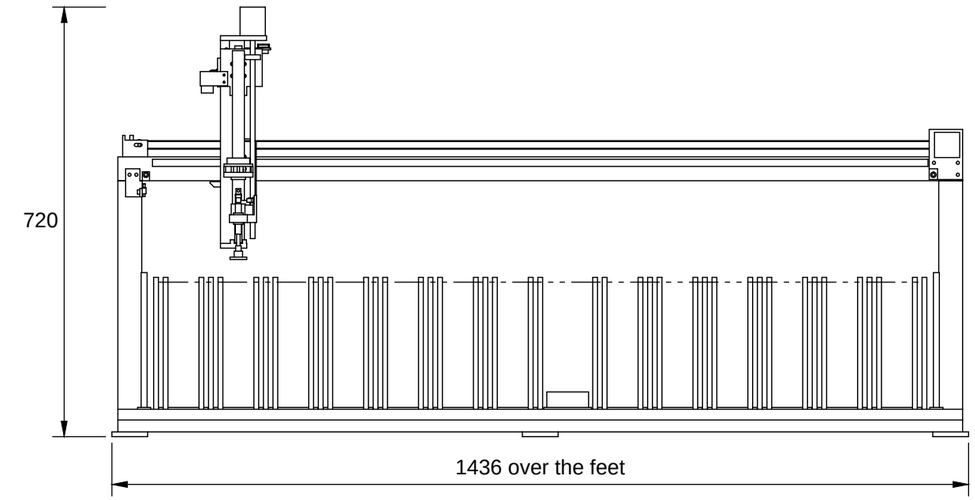
TOP

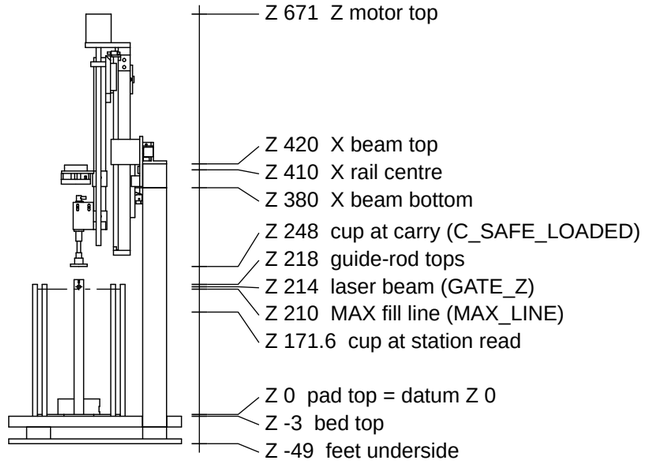


ISO (render, docs/img/machine_iso.png; not to scale)





FRONT (looking along +Y)



RIGHT SIDE (along -X)

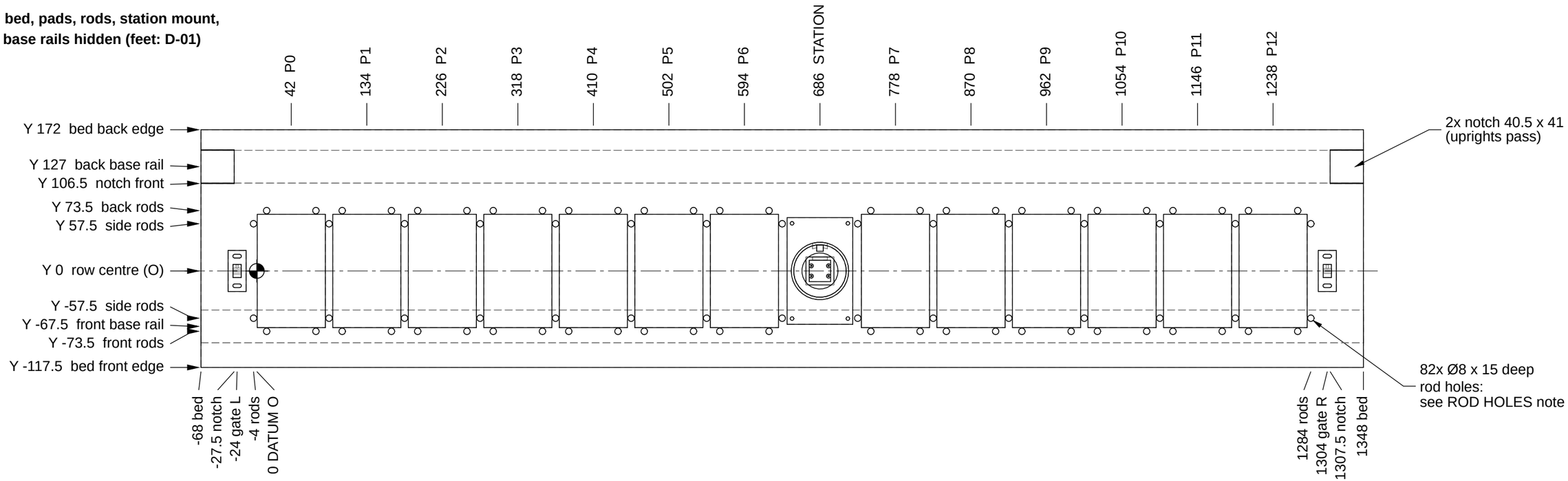
Heights, machine Z (cup face where noted)

Main data	Value	From
Pockets, one row	13 + station (cell 7)	layout
Pitch; slabs per pocket	92; 30	layout
Pocket interior X x Y	84 x 139	layout
Guide rods	82 x Ø8	CAD
X used (cup X)	42 to 1238	CAD
X stops; home trip	15 / 1262; 19	layout
Z used (cup face c)	-12 to 250	layout
Z hard stops (c)	-14 / 262	layout

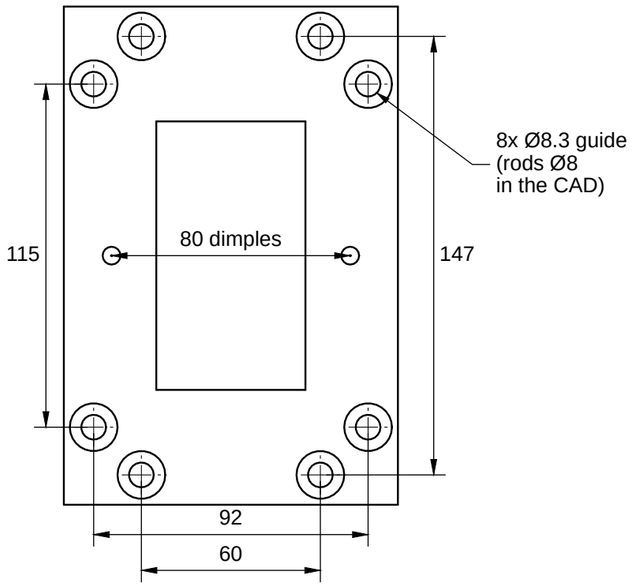
Pose 'home' (build.POSES): cup over pocket 1 at the carry height. Not shown: fasteners, slabs, light roof, cables. Heights are machine Z above the pad top (Z 0); X from the machine origin O (D-03). Dash-dot: MAX fill line; cross on the right view: the overfill laser beam.

SlabRail - graded-slab batch scanner	**SHEET D-01 of D-17**
General arrangement	SCALE 1:10
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ c50a3c7 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

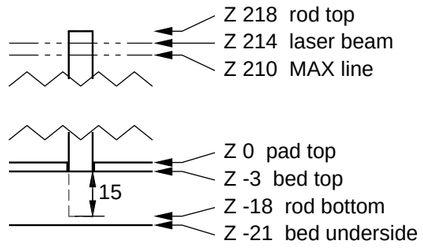
PLAN (1:5): bed, pads, rods, station mount, gate posts; base rails hidden (feet: D-01)



DETAIL C (1:2): one cell's rod holes = drill_template
(cell centre at 0, 0; the machine drills them, BUILD_GUIDE 5.6)



DETAIL D (1:2): rod in the bed,
front, broken



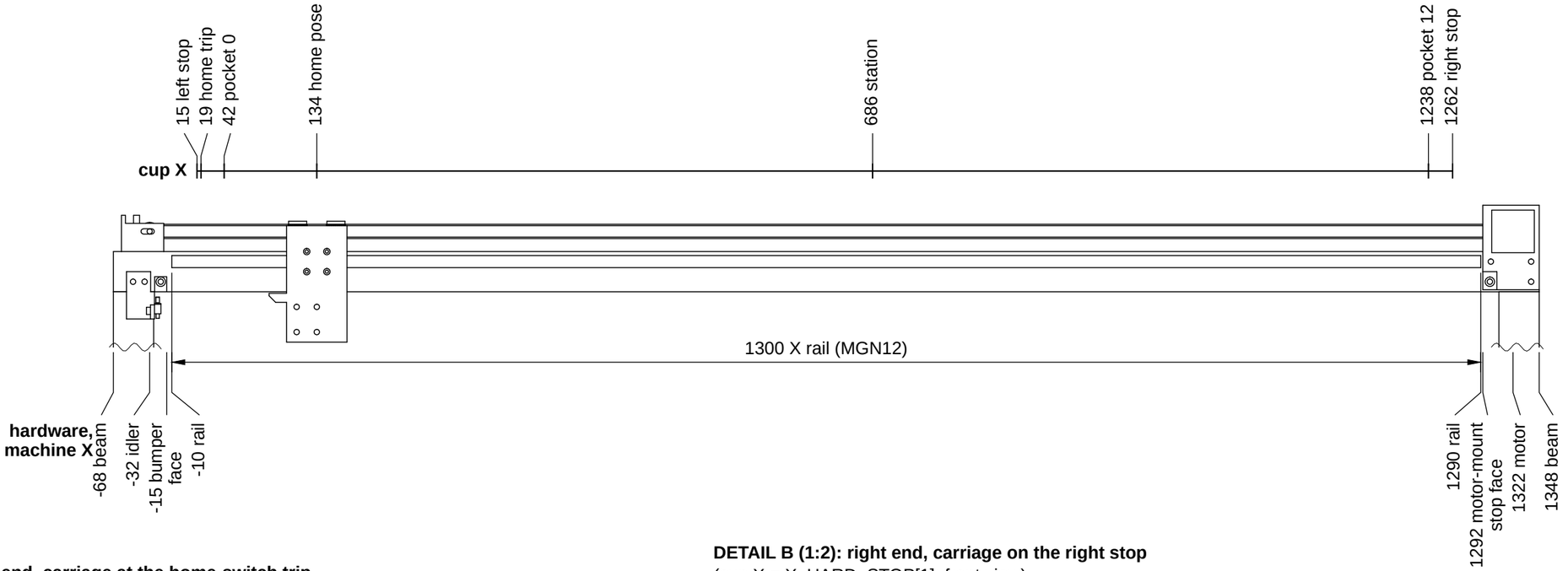
Bed and pads

Item	Size (from the CAD)	Note
Bed, plywood (W)	1416 x 289.5 x 18	cut on site; test-drill for voids first (BUILD_GUIDE 5.1)
Pads (EVA + microfiber), 13	83 x 138 x 3	spray adhesive, after the rods (5.9)
Rods (dowels), 82	D-02 item C-11	glued; plumb within 1 over 200 [Guessing target] (5.7)

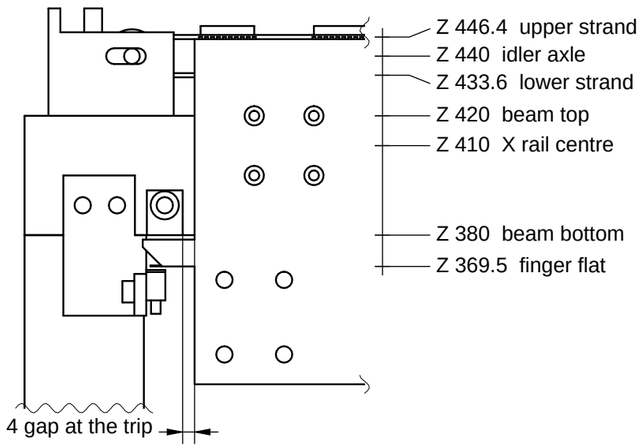
Datum O = machine origin (X 0; Y 0 = the pocket-row centre line), as the software uses it; X ordinates from O. The machine drills the rod holes through drill_template, so they are not laid out from the bed edges (BUILD_GUIDE 5.6). Station mount: cell 7 (X 686), four #8 wood screws, no pad. Gate posts: bare plywood, no shim (5.9).

ROD HOLES: 82x Ø8 (ROD_D; brad-point bit to suit the dowels: 5/16 in for the BOM's 7.94 dowels) x 15 deep (ROD_EMBED), square to the bed (guide block, 5.6). Side rods on the wall lines: X -4, then every 92, at Y ±57.5. Front/back rods: pairs 60 apart about each pocket centre, at Y ±73.5; none in cell 7.

SlabRail - graded-slab batch scanner	**SHEET D-03 of D-17**
Bed, pockets and guide rods	SCALE 1:2, 1:5 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ c50a3c7 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**



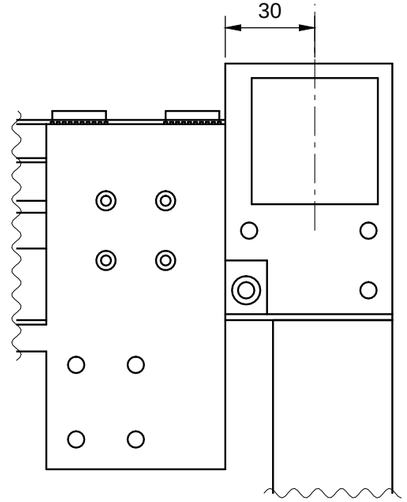
DETAIL A (1:2): left end, carriage at the home-switch trip
(cup X = X_HOME_SWITCH; front view)



Click at cup X 19 (+/-2, BUILD_GUIDE 6.8)

FRONT: X axis with the carriage at home (cup over pocket 1); Z axis and head not shown. cup X = machine X of the suction-cup axis. Belt: GT2 loop over the beam top, both ends clamped on the x_bracket tab (BUILD_GUIDE 6.5).

DETAIL B (1:2): right end, carriage on the right stop
(cup X = X_HARD_STOP[1]; front view)



x_bracket lands on the mount's left end

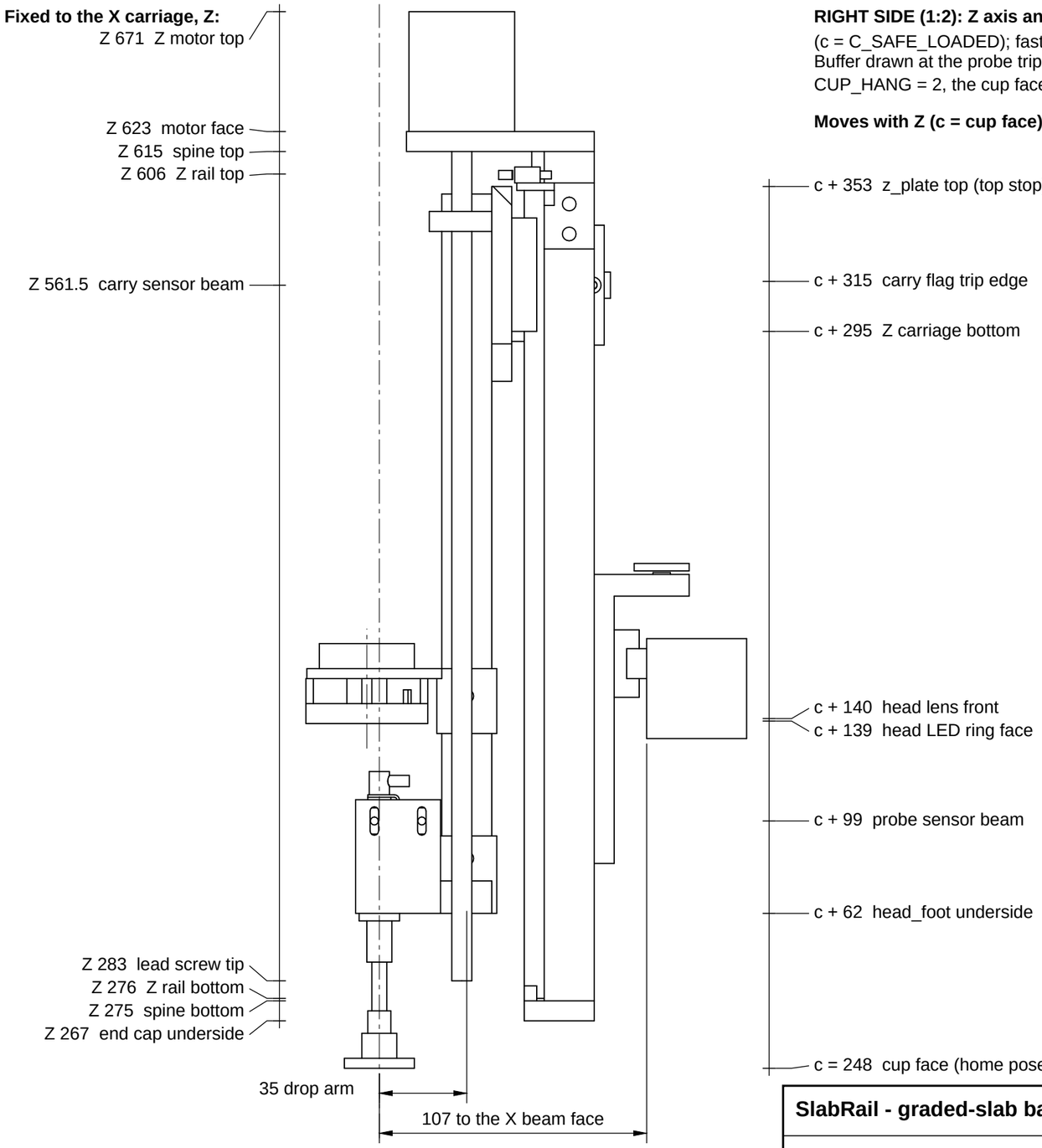
Y stack-up, cup axis to the X beam's front face

Constant (layout.py)	mm	Part
ARM_TO_CUP	35	drop-arm centre to cup axis
EXT20 / 2	10 (of 20)	drop arm (2020): half = 10
Z_PLATE_T	8	z_plate
MGN12_H	13	Z carriage (MGN12H)
EXT20	20	Z spine (2040, 20 deep)
X_BRACKET_T	8	x_bracket
MGN12_H	13	X carriage (MGN12H)
= Y_CUP_TO_BEAM	107	sum: cup axis to beam front face

CAD check: beam front face at Y 107

SlabRail - graded-slab batch scanner	**SHEET D-04 of D-17**
X axis	SCALE 1:2, 1:5 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ c50a3c7 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

Fixed to the X carriage, Z:



RIGHT SIDE (1:2): Z axis and head, home pose

(c = C_SAFE_LOADED); fasteners not shown
Buffer drawn at the probe trip: compressed
CUP_HANG = 2, the cup face at c.

Moves with Z (c = cup face):

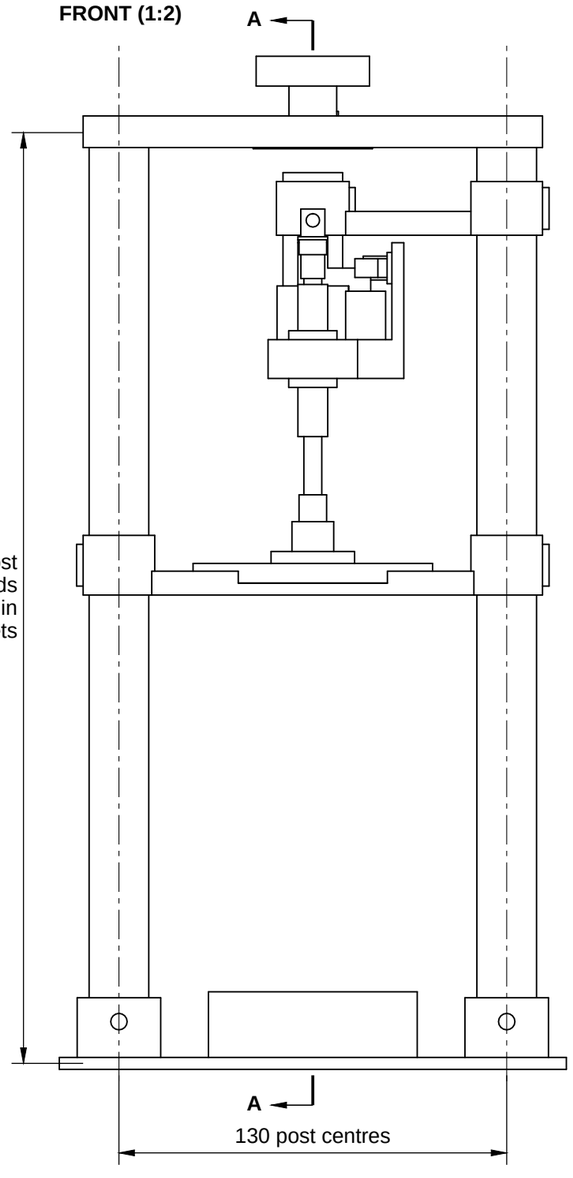
Z-height table (values read from cad/layout.py)

Constant	Value	Meaning
SLAB_T	6.6	c at pick contact = top of the stack: one slab (pose pick_bottom) = SLAB_T; n slabs = n x SLAB_T, nominal
CUP_HANG	2	the free cup hangs this far below c; the probe trips after this much buffer travel
PROBE_TRIP_TOL	0.5	trip tolerance on CUP_HANG (BUILD_GUIDE 7.6)
PROBE_WINDOW	12	max probing travel past the predicted contact
BUFFER_STROKE	20	spring buffer stroke (SMC ZPT25UUK20-B5-A10)
STATION_LENS_Z	23	station lens front above the pad top
STATION_READ_C	171.6	c at the station read = STATION_LENS_Z + CAM_TO_SLAB + SLAB_T + CUP_HANG
CAM_TO_SLAB	140	both cameras: lens front to the slab face
HEAD_CAM_DY	-5	head lens axis, Y from the cup axis (negative = toward the label end)
POST_H	218	guide-rod tops above the pad top
C_SAFE_LOADED	248	carry height, loaded and empty (C_SAFE_EMPTY is the same)
CARRY_SENSOR_C0	246.5	carry sensor: beam blocked from this c up to the top stop
CARRY_BAND	1	carry sensor band: within it either reading is allowed
C_MAX	250	top of the software's Z travel
Z_SWITCH_C	258	Z top (home) switch trip
C_HARD_TOP	262	top hard stop: z_plate meets z_motor_mount
C_MIN	-12	lowest c the design allows
C_HARD_BOTTOM	-14	bottom hard stop: Z carriage skirts on the z_end_cap lips
Z_CARRIAGE_ABOVE_CUP	295	Z carriage bottom edge above the cup face
Z_SPINE_BOTTOM	275	spine bottom end above the pad top (set +/-1, BUILD_GUIDE 7.1)
Z_SCREW_DX	24	lead screw axis, X from the cup axis
Z_SPINE_OFFSET_X	-10	Z spine centre, X from the cup axis

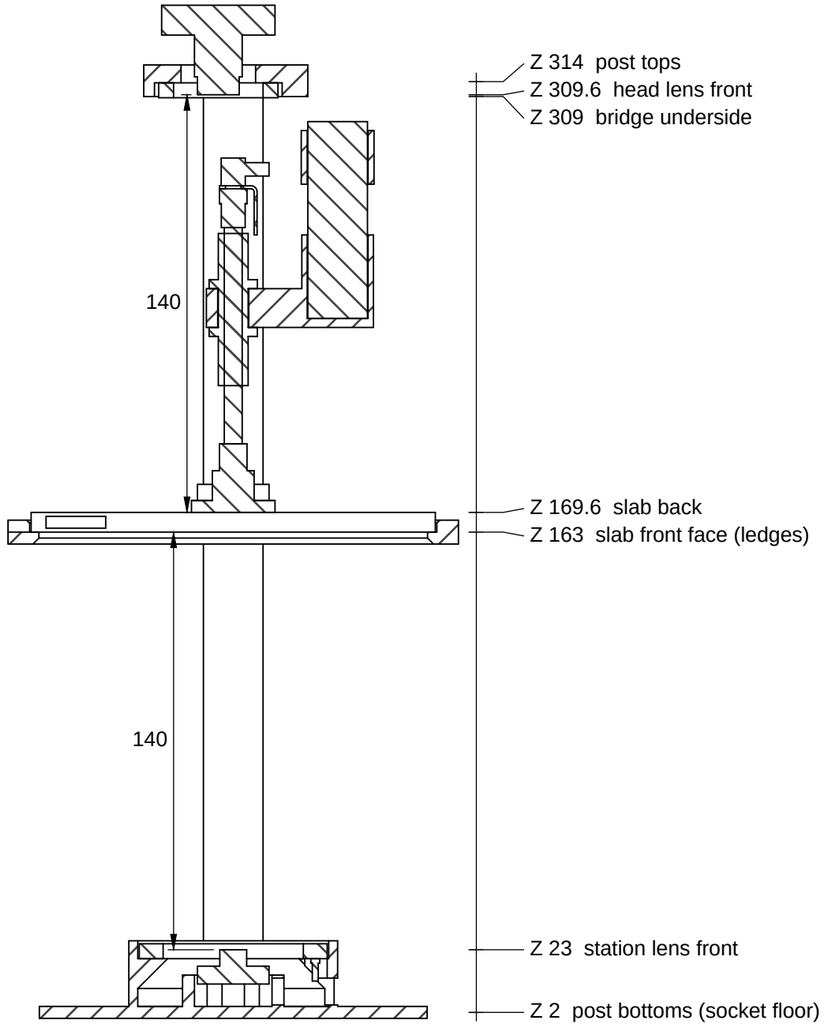
SlabRail - graded-slab batch scanner	**SHEET D-05 of D-17**
Z axis and head stack-up	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ c50a3c7 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

PHASE 0A TEST JIG: one slab between the two cameras, as on the machine (BUILD_GUIDE 1.5); fasteners not shown

FRONT (1:2)

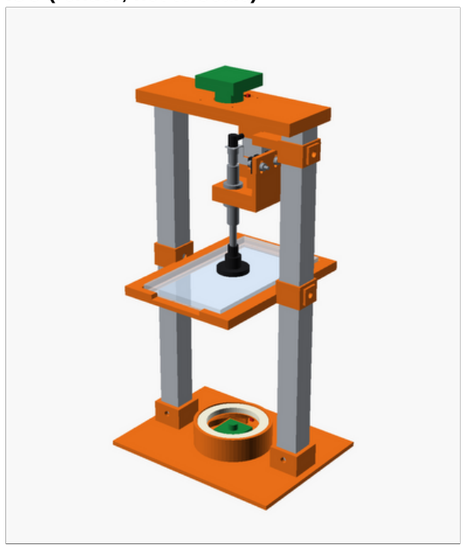


SECTION A-A (1:2)
on both lens axes, seen from +X; cut faces hatched



Heights above the base underside

ISO (render; not to scale)



Part	Type	Qty	Sheet
phase0_base (camera mast, ring seat, post sockets)	P	1	D-15
phase0_tray (the slab on edge ledges)	P	1	D-15
phase0_top (bridge: head camera, head ring)	P	1	D-16
phase0_occluder_mount	P	1	D-16
head_foot, the machine's own (occluder)	P	1	D-12
2020 post, 312, cut square +/-0.5 (C-08)	C	2	D-02
2020 arm stub, 66 (C-09)	C	1	D-02
probe flag (C-07)	S	1	D-06
cameras, LED rings, SMC buffer + cup + elbow, endstop board	B	-	BOM.csv

The posts set the bridge height: cut them square. Both lens-to-slab distances (SECTION A-A) equal CAM_TO_SLAB = 140 on the machine (set +/-2, BUILD_GUIDE 1.5). Phase 0 is the go / no-go test before any motion parts are bought.

SlabRail - graded-slab batch scanner	**SHEET D-17 of D-17**
Phase 0 test jig	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ c50a3c7 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**